INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA
Term – I ,Third Semester Examination, September- 2026
Subject Code: BCACC301
Subject Name: Design and Analysis of Algorithms

Full Marks: 30 Set No: 2 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2×5)

1. What are the two main components considered while analyzing the efficiency of an algorithm? **2**

2. Differentiate between Big-O and Big-Ω notation. **2**

3. Arrange the following complexities in increasing order of growth:

$O(n^2)$, O(1), O(n log n), O(log n), O(n)

4. What is the best-case time complexity of Bubble Sort? Under what condition is it achieved? **2**

5. For the array A= [5, 2, 4, 6] what will be the array after the first pass of Bubble Sort? **2**

6. Write the recurrence relation for the divide-and-conquer maximum-minimum algorithm. **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5×2)

7. Explain best-case, average-case and worst-case analysis using linear search as an example. **5**

8. Trace binary search to find 15 in the following sorted array: **5**

[5, 10, 15, 20, 25, 30, 35, 40, 45]

Show the values of `low`, `high` and `mid` at every step.

9. Consider the array: **5**

A=[29, 10, 14, 37, 13]

Perform Insertion Sort on the above array in ascending order. Show the array after each major iteration and state the worst-case time complexity of Insertion Sort.

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10×1)

10. Explain Big-O, Big-Omega and Big-Theta notations mathematically and graphically. **10**

Determine suitable asymptotic bounds for:

$f_1(n) = 4n + 10$
$f_2(n) = 6n^2 + 3n + 8$
$f_4(n) = n \log n + 10n$

11. Illustrate Merge Sort with example and perform step-by-step complexity analysis. **10**

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